

(12) **United States Patent**
Miller et al.

(10) **Patent No.:** **US 12,404,669 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **CONSTRUCTION USING BUNDLED TUBE
AND THREADED STEPPED DOWELS**

(71) Applicant: **Miller Dowel Company**, Wilmette, IL
(US)

(72) Inventors: **Michael R. Miller**, Winnetka, IL (US);
Robert Daly, Elk Grove Village, IL
(US)

(73) Assignee: **Miller Dowel Company**, Wilmette, IL
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 133 days.

(21) Appl. No.: **18/174,432**

(22) Filed: **Feb. 24, 2023**

(65) **Prior Publication Data**

US 2023/0220660 A1 Jul. 13, 2023

Related U.S. Application Data

(60) Continuation of application No. 17/811,264, filed on
Jul. 7, 2022, now Pat. No. 11,613,885, which is a
(Continued)

(51) **Int. Cl.**
E04B 1/26 (2006.01)
E04B 1/48 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **E04B 1/2604** (2013.01); **E04B 1/486**
(2013.01); **E04C 3/36** (2013.01); **F16B 13/02**
(2013.01); **E04B 2001/2692** (2013.01)

(58) **Field of Classification Search**

CPC F16B 13/02; F16B 12/24; Y10T 403/559;
Y10T 403/55; A47B 2230/0037;
(Continued)

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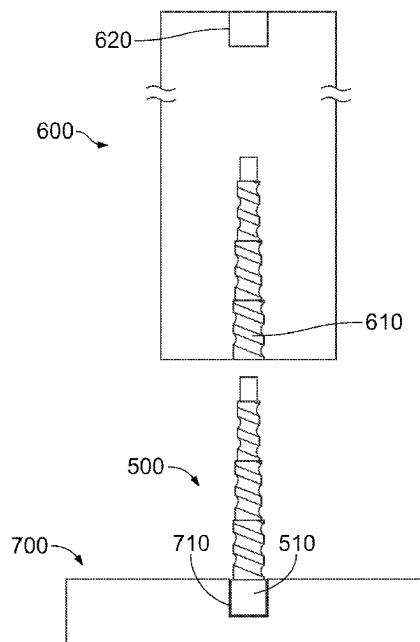
Primary Examiner — Theodore V Adamos

(74) *Attorney, Agent, or Firm* — Husch Blackwell LLP

(57) **ABSTRACT**

A system and method for creating a column for construction using bundled tube are provided. The column having a bottom portion, a middle portion, and a top portion. Each portion consists of a bundle of individual pillars, comprising a center pillar, a plurality of corner pillars, and a plurality of middle pillars. From the bottom portion of the column, addition pillars can be affixed on top of the pillars of the bottom portion, thus forming the middle and the top portion of the column. Each pillar can be secured to another component using a thread stepped dowel comprising of a base section, at least one middle section, and a tip section, with each section being progressively smaller in circumference than the last. The at least one middle section of the threaded stepped dowel further comprises thread spiraling around the outer surface of the at least one middle section.

18 Claims, 8 Drawing Sheets



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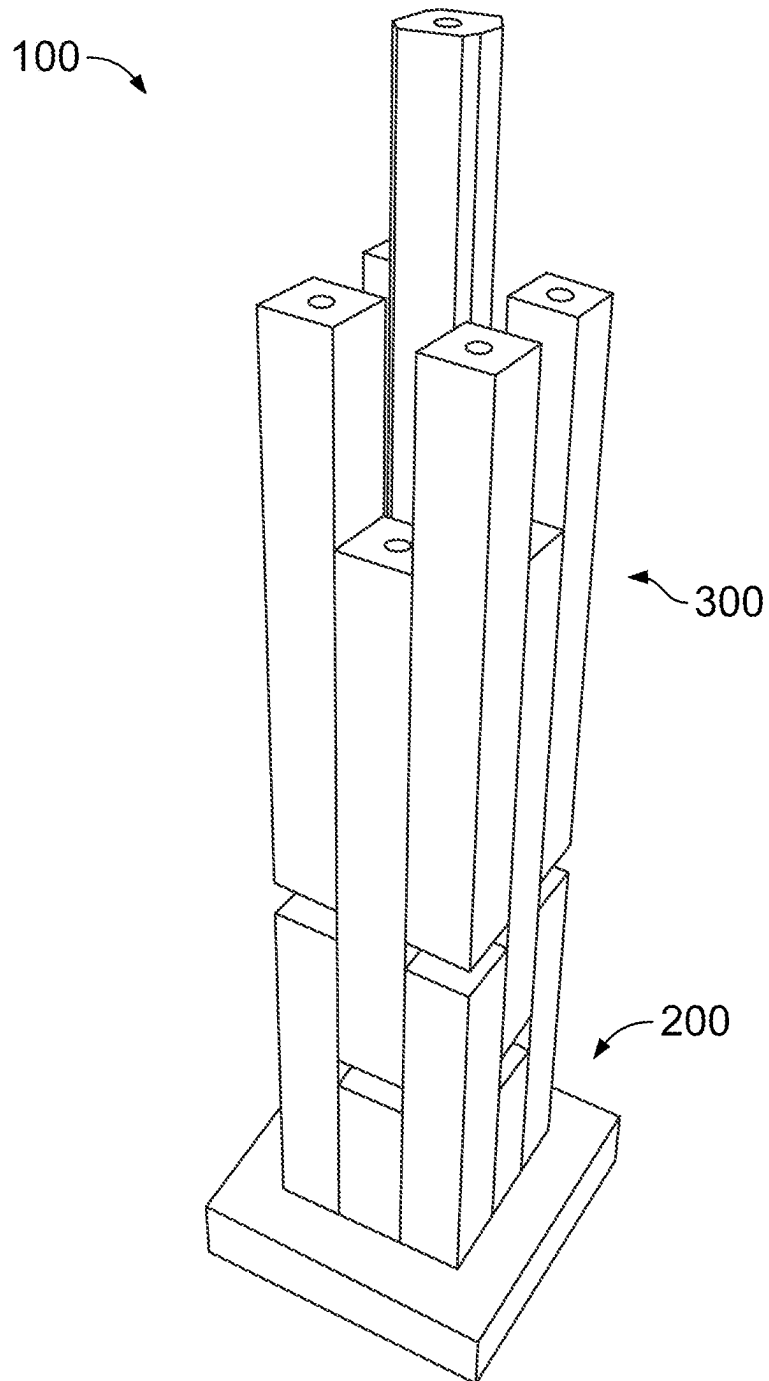


FIG. 1

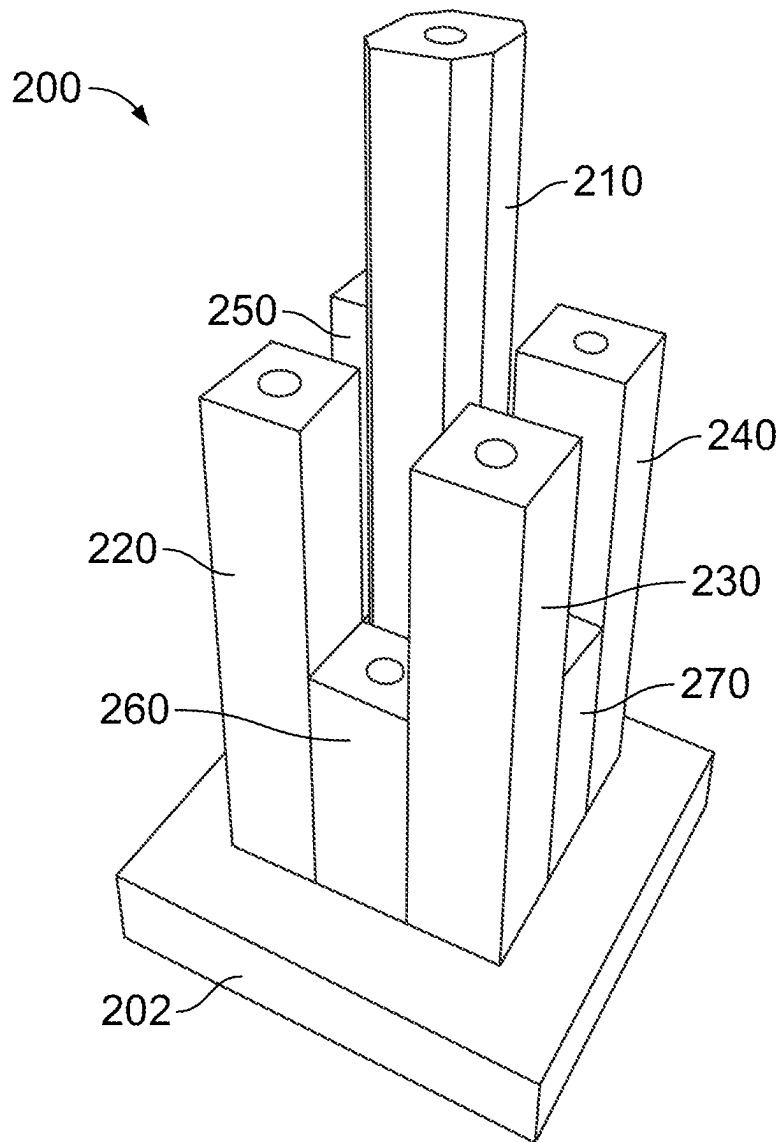


FIG. 2

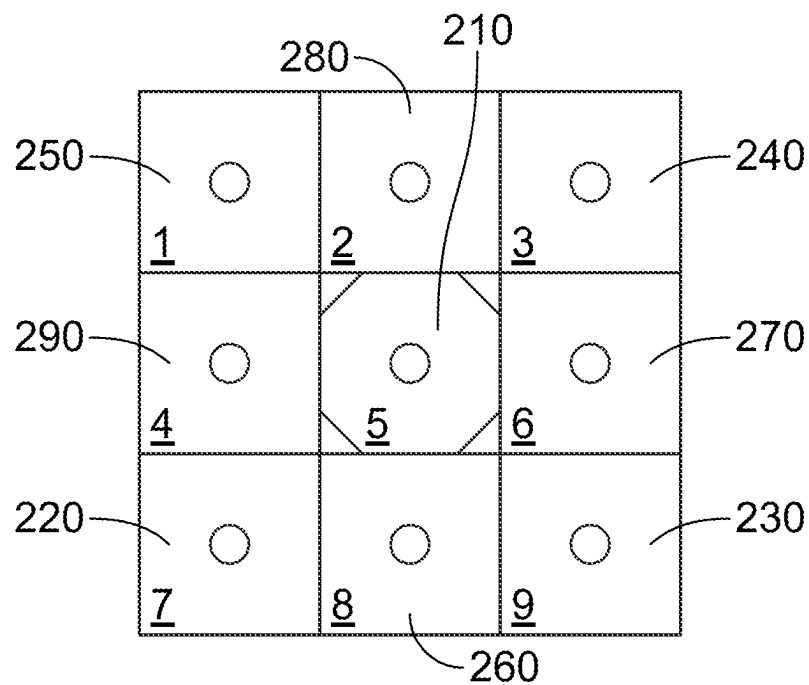


FIG. 3

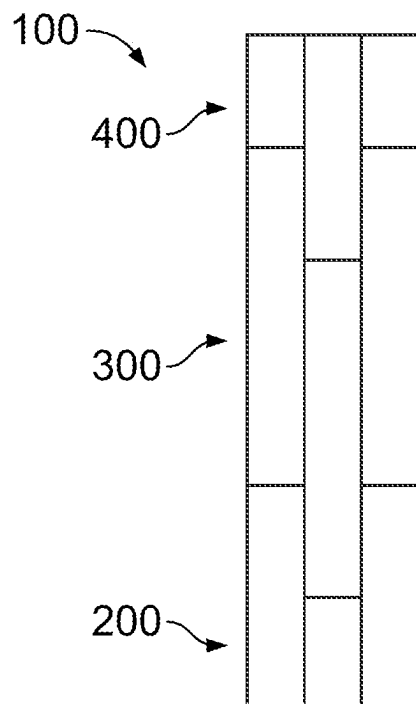


FIG. 4

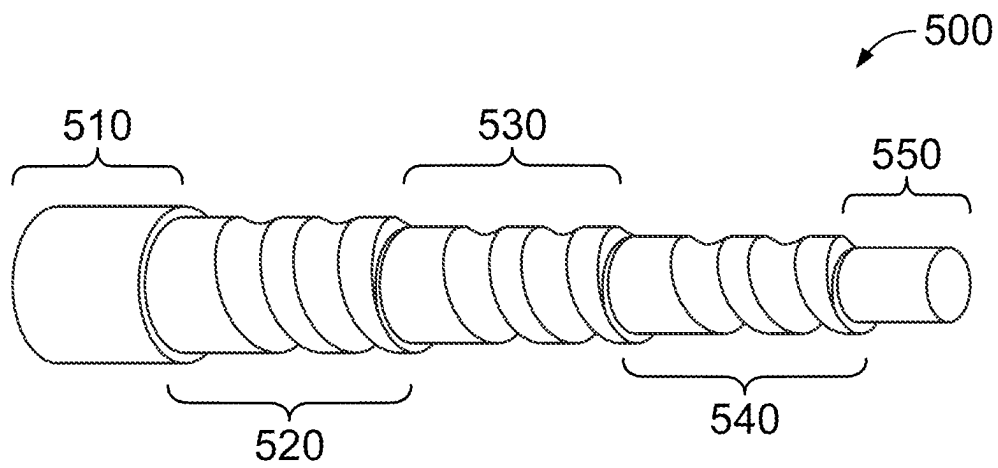


FIG. 5A

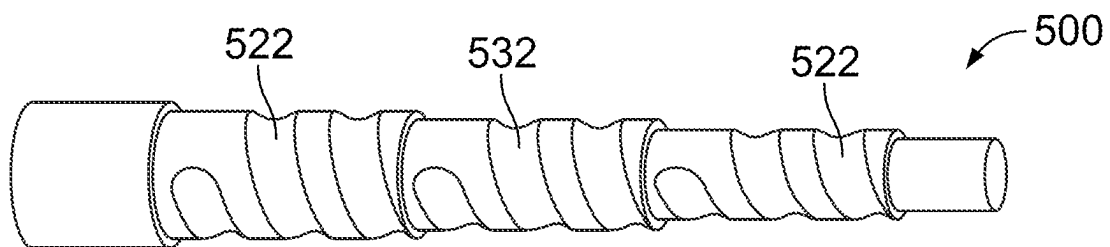


FIG. 5B

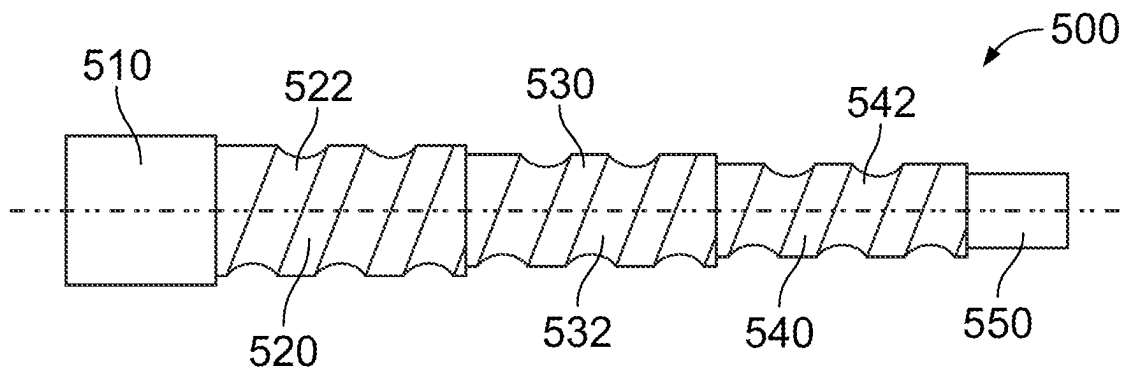


FIG. 6A

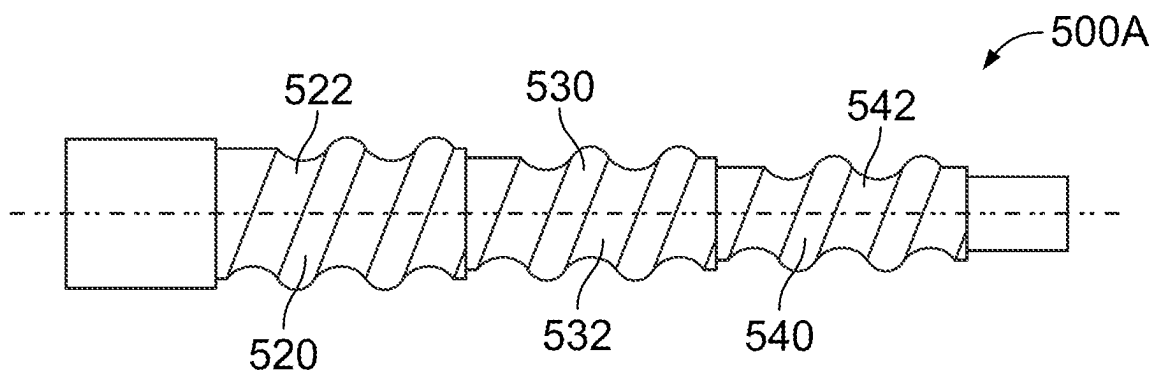


FIG. 6B

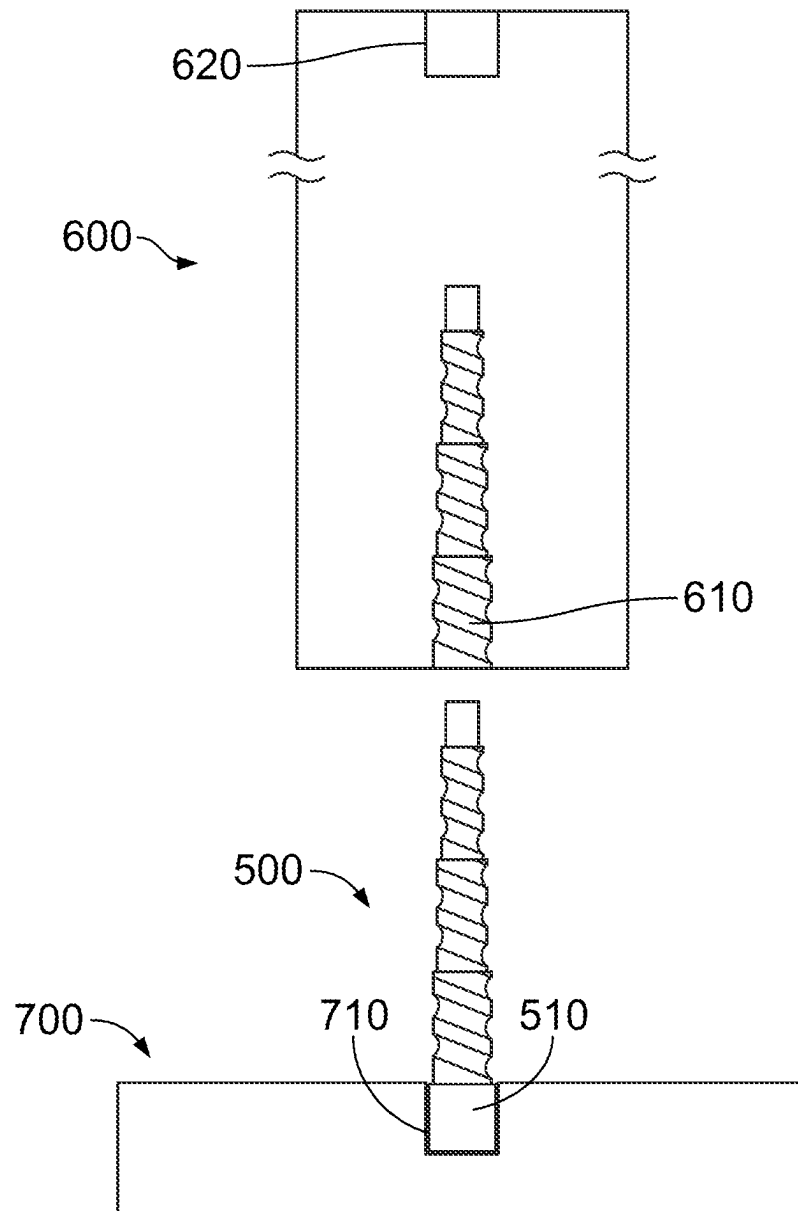


FIG. 7

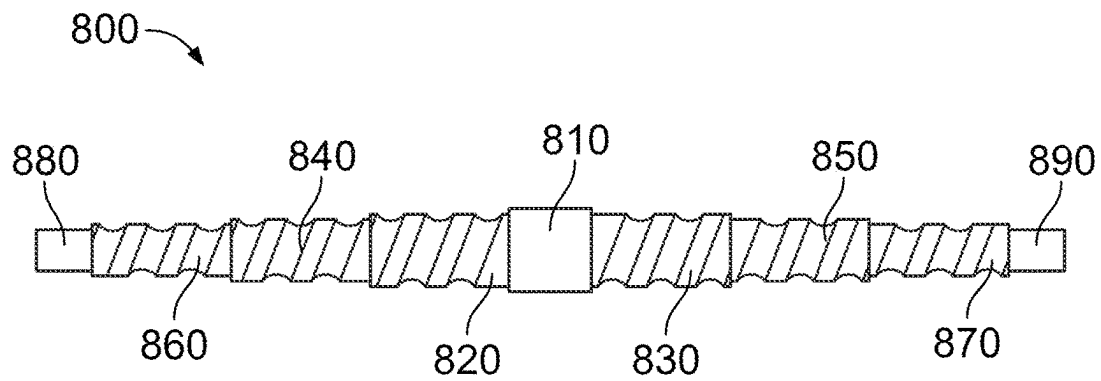


FIG. 8A



FIG. 8B

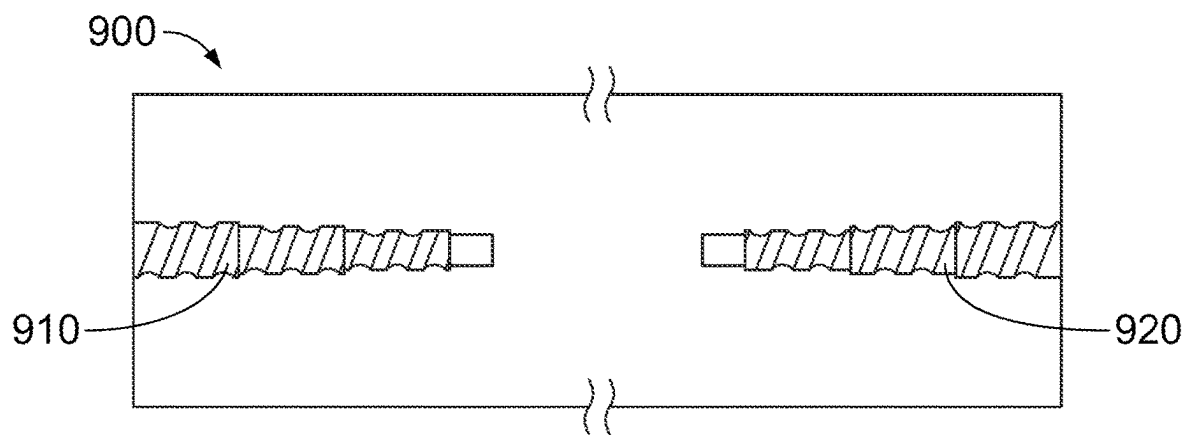


FIG. 9

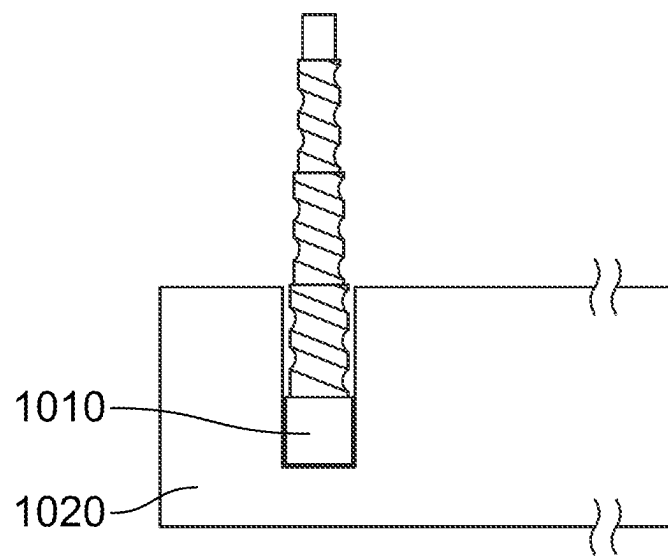
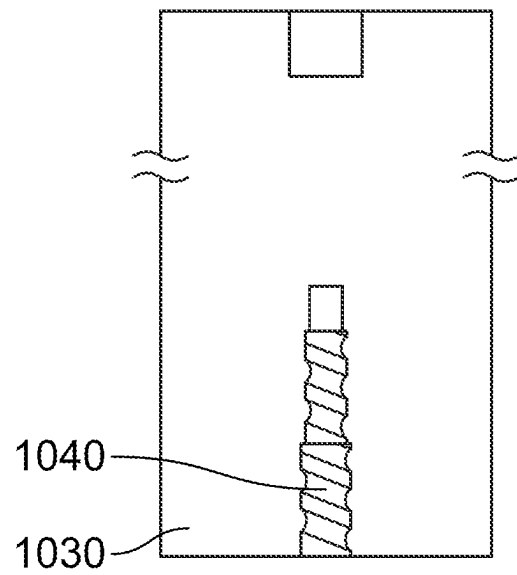


FIG. 10

CONSTRUCTION USING BUNDLED TUBE AND THREADED STEPPED DOWELS

CROSS-REFERENCE TO RELATED APPLICATION

This application is continuation of U.S. application Ser. No. 17/811,264 filed Jul. 7, 2022, entitled “Construction Using Bundled Tube And Threaded Stepped Dowels”, which is a divisional application of U.S. application Ser. No. 17/158,933 filed Jan. 26, 2021, entitled “Construction Using Bundled Tube And Threaded Stepped Dowels”, which claims the benefit of the filing date of U.S. Provisional Application Ser. No. 62/966,405 filed Jan. 27, 2020, entitled “Improved Apparatus And Method For Assembly Of Construction Components” and U.S. Provisional Application Ser. No. 63/057,399 filed on Jul. 28, 2020, entitled “Construction Using Bundled Tube And Threaded Stepped Dowels”, all of which are hereby incorporated by reference as if fully set forth herein.

BRIEF SUMMARY OF THE INVENTION

This disclosure generally relates to using bundled tube to form columns and beams for construction. The disclosure also relates to threaded stepped dowels for enhanced connection of adjoining construction components and to a method of assembling the bundled tube using the threaded stepped dowel to form a desired configuration.

BACKGROUND OF THE INVENTION

Wood has been used as a construction material for more than a thousand years. For example, Horyu-ji Temple in Japan is believed to be the oldest wooden building in the world, which was built more than 1,300 years ago.

However, for most of the twentieth century, constructions of buildings have mainly relied on utilizing reinforced concrete, such as post-tensioned concrete. In contrast, buildings built out of timber or engineered wood were relatively few and far in between. This began to change since the development of cross-laminated timber (CLT) in early 1990s.

CLT is a multilayer solid wood panel often referred to as thick wood or cross-laminated wood. Such CLT panels form solid wood panels that can be used in construction. CLT usually consists of several flat overlying flat board layers, unlike glued laminated timber (glulam) in which the layers are arranged longitudinally to the fiber.

CLT panel construction is the next level in quality and speed of construction. Wall, floor, and roof elements manufactured in a climate-controlled facility and transported to building sites for rapid assembly dramatically increases quality control in the building process.

Today, CLT is widely used in residential and light engineering structures in situations where large beam depths are required, such as long span openings in houses. However, using CLT to build skyscrapers and other megastructures remain elusive in the industry.

Several methods of construction are common. The first is known as platform construction or endoskeleton construction. Platform construction is the primary method utilized in the United States. In essence, floor joists rest on a sill plate or on top of a stud wall. The next level of wall framing will then sit on top of the fully sheathed floor joists.

Although platform construction can be utilized for smaller structures, building taller buildings out of platform construc-

tion is not ideal. Specifically, with platform construction, each additional level rests on top of the level below, meaning the weight of the higher floors is bear by the lower floors.

Unlike vertical posts and pillars, floor and ceiling panels generally consist of side-grain wood, which only has a crushing strength of around 500 pounds per square inch (psi). On the other hand, vertical pillars consist mostly of end-grain wood, which has a crushing strength of about 5,000 to 7,000 psi for most species of wood.

Another issue with platform construction is the likelihood of compression. Mismatching floor and ceiling panels that consist mostly of side-grain wood with pillars consist mostly of end-grain wood permits a wider latitude of compression, which can be damaging for a structure.

In contrast, a second method is known as balloon construction or exoskeleton construction. Although once prevalent in the United States, this construction method is now more popular in Europe. Unlike a platform construction, the wall stud rests on the sill plate with a rim joist in the interior side and then the floor joist. The stud wall is continuous from the sill plate to the top plate. At the second level the floor connection, joists rest on a ledger and are then face nailed to the studs. Thus, there is a need to create a continuous column or stud tall enough for the construction of a superstructure.

An issue with building a skyscraper out of wood is the need of large support columns that are also made of wood. Creating a column that can withstand the weight of a tall building is engineeringly challenging. Thus, there is a need for a column, made of wood, that can be used in constructing larger wooden buildings.

Likewise, joining wood construction components remain a constant challenge in the industry. Traditionally, metal fasteners such as screws, nails, or rods have been used to join together adjoining wooden components. However, the materialistic property of metal is fundamentally different from that of wood. The problem is further amplified when attempting to join together large pieces of construction components, such as walls, roofs, beams, and columns. Thus, there is a need in for a mechanism to join wooden construction materials using a wooden fastener.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a perspective view of a bundled tube according to an exemplary embodiment.

FIG. 2 illustrates another perspective view of the bundled tube according to the exemplary embodiment.

FIG. 3 illustrates a top view of the bundled tube according to the exemplary embodiment.

FIG. 4 illustrates a side view of the bundled tube according to the exemplary embodiment.

FIGS. 5A and 5B illustrate perspective views of a threaded stepped dowel according to an exemplary embodiment.

FIG. 6A illustrates a side view of the threaded stepped dowel according to the exemplary embodiment.

FIG. 6B illustrates a side view of a threaded stepped dowel according to another exemplary embodiment.

FIG. 7 illustrates a cross sectional view of a first component to be affixed onto a second component using a threaded stepped dowel according to an exemplary embodiment.

FIGS. 8A and 8B illustrate side views of alternative designs of a threaded stepped dowel according to another exemplary embodiments.

FIG. 9 illustrates a cross sectional view of a pillar according to another embodiment.

FIG. 10 illustrates a cross sectional view of attaching two components with a threaded stepped dowel according to an embodiment.

Before explaining the disclosed embodiment of the present invention in detail, it is to be understood that the invention is not limited in its application to the details of the particular arrangement shown, since the invention is capable of other embodiments. Exemplary embodiments are illustrated in referenced figures of the drawings. It is intended that the embodiments and figures disclosed herein are to be considered illustrative rather than limiting. Also, the terminology used herein is for the purpose of description and not of limitation.

DETAILED DESCRIPTION

While this invention is susceptible of embodiments in many different forms, there are shown in the drawings and will be described in detail herein specific embodiments with the understanding that the present disclosure is an exemplification of the principles of the invention. It is not intended to limit the invention to the specific illustrated embodiments. The features of the invention disclosed herein in the description, drawings, and claims can be significant, both individually and in any desired combinations, for the operation of the invention in its various embodiments. Features from one embodiment can be used in other embodiments of the invention.

As shown in FIGS. 1-9, the embodiments of this disclosure include a threaded stepped dowel, and a column formed from a bundled tube.

In order to construct taller buildings and larger structures using engineered wood such as CLT, especially in circumstances where balloon construction is preferred, a new way to form a larger column or stud is necessary. This is due in part to the difficulty of transporting one massive pre-constructed column from a prefabrication plant, but also the difficulty of creating a column large enough to support a superstructure off-site.

FIG. 1 shows a novel way to construct a column using bundled tube. Instead of having one massive column made from wood, the column 100 is created by combining a plurality of smaller pillars of different lengths together to form the larger column 100. Depending on the need of the specific building, the column 100 can range from several feet tall to hundreds of feet tall. That is to say, using the principle disclosed herein, the length of the column 100 can vary to suit its purpose. A column 100 can include a base portion 200, a middle portion 300, and a top portion 400.

Referring to FIGS. 2 and 3, the base portion 200 of the column 100 is formed by at least nine individual pillars in a three by three layout. A sill plate or a base 202 is provided at the bottom most portion of a building. From there, nine pillars are utilized to build the base portion 200.

First, a center pillar 210 is affixed onto the base 202 in the center of a three-by-three grid. I.e., position 5 as shown in FIG. 3. The center pillar 210 is an elongated pillar with a top surface and a bottom surface. In an embodiment, the center pillar 210 can be made out of engineered wood such as CLT. In another embodiment, the center pillar 210 can be made out of natural timber.

A first bore is provided at a center portion of the bottom surface of the center pillar 210, and a second bore is provided at a center portion of the top surface of the center pillar 210. The first bore and the second bore are used to receive fasteners so that the center pillar 210 can be affixed

on top of the base 202, and that another center pillar can be affixed on top of the center pillar 210.

In an embodiment, the first bore and the second bore can be separate and distinct, i.e., they do not bore through the entire center pillar 210. In another embodiment, the first bore and the second bore can connect, thus forming one continuous bore through the center pillar 210. The internal shapes of the first bore and the second bore varies depending on the type of fastener being used. That is to say, the first bore and the second bore need be the same shape.

The center pillar 210 can be affixed to the base 202 through a variety of means. In an embodiment, the center pillar 210 is affixed to the base 202 through a stepped dowel such as the ones disclosed in U.S. Pat. No. 6,871,681, which is incorporated by reference in its entirety herein. In another embodiment, the center pillar 210 can be twisted onto the base 202 by using a threaded stepped dowel that will be described in more detailed later. Other fasteners can also be used, such as metal studs or rods known in the art.

The fastener is affixed onto the base 202 through any appropriate mean. Thereafter, the fastener, which is protruding out of the base 202, is received in the first bore of the center pillar 210, which is located at the bottom surface of the center pillar 210.

In an embodiment, as seen in FIGS. 2 and 3, the center pillar 210, when viewed from top down, is octagonal in shape to provide space for corner pillars 220, 230, 240, 250, to be twisted into place. In another embodiment, the center pillar 210 can be cylindrical. In yet another embodiment where the corner pillars 220, 230, 240, 250, need not be twisted into place, such as in cases where the corner pillars 220, 230, 240, 250, are dropped in place instead, the center pillar 210 can be rectangular in shape. Other shapes can also be possible depending on the specific circumstances.

Once the center pillar 210 is affixed onto the base 202, the corner pillars 220, 230, 240, 250 are then affixed onto the base 202. Similar to the center pillar 210, the corner pillars 220, 230, 240, 250 are elongated pillars with respective top surfaces and bottom surfaces. The corner pillars 220, 230, 240, 250, are generally rectangular in shape when viewed top down. In an embodiment, the corner pillars 220, 230, 240, 250 are shorter than the center pillar 210 as shown in FIG. 2. In another embodiment, the corner pillars 220, 230, 240, 250 can be longer than the center pillar 210. In yet another embodiment, the corner pillars 220, 230, 240, 250 can be the same length as the center pillar 210. It is preferred that the corner pillars 220, 230, 240, 250 have the same length as one another, but each individual corner pillar can have different lengths if necessary.

Similar to the center pillar 210, each of the corner pillars 220, 230, 240, 250 has a first bore on its respective bottom surface, and a second bore on its respective top surface, each extending inward from their respective surfaces. These bores are used to affix the corner pillars 220, 230, 240, 250 to the base 202 from the bottom, and to affix additional corner pillars on top of the corner pillars 220, 230, 240, 250.

In an embodiment, threaded stepped dowels can be used to affix the corner pillars 220, 230, 240, 250 to the base 202 to, referring to FIG. 3, positions 7, 9, 3, and 1 respectively. When threaded stepped dowels are used, the corner pillars 220, 230, 240, 250 can be twisted onto the base 202 during installation, thus affixing the corner pillars 220, 230, 240, 250 onto the base 202. To accommodate for the twisting motion during installation, it is preferred that the center pillar 210 is octagonal as described above, although the center pillar 210 can also be cylindrical or other appropriate shapes.

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In another embodiment, the corner pillars **220, 230, 240, 250** can be dropped onto fasteners protruding out of the base **202** instead of being twisted onto the base **202**. In this embodiment, un-threaded stepped dowels can be used, as well as other types of conventional fasteners known in the art.

Once the corner pillars **220, 230, 240, 250** are affixed onto the base **202** in addition to the center pillar **210**, middle pillars **260, 270, 280, 290** are then affixed onto the base **202**. As with other pillars, the middle pillars **260, 270, 280, 290** are elongated pillars each having a top surface and a bottom surface. The middle pillars **260, 270, 280, 290** are generally rectangular in shape, although other shapes can be possible.

Again, each of the middle pillars **260, 270, 280, 290** has a first bore on its respective bottom surface, and a second bore on its respective top surface, each extending inward from their respective surfaces. These bores are used to affix the middle pillars **260, 270, 280, 290** to the base **202** from the bottom, and to affix additional middle pillars on top of the middle pillars **260, 270, 280, 290**.

In an embodiment, the middle pillars **260, 270, 280, 290** are shorter than the center pillar **210** and also short than the corner pillars **220, 230, 240, 250** as shown in FIG. 2. In another embodiment, the middle pillars **260, 270, 280, 290** can be longer than the center pillar **210** or the corner pillars **220, 230, 240, 250** or both. In yet another embodiment, the middle pillars **260, 270, 280, 290** can be the same length as the center pillar **210** or the corner pillars **220, 230, 240, 250** or both. It is preferred that the middle pillars **260, 270, 280, 290** have the same length as one another, but each individual middle pillar can have different lengths if necessary.

During installation, the middle pillars **260, 270, 280, 290** are dropped onto fasteners protruding out of the base **202** at, referring to FIG. 3, positions **8, 6, 2, and 4** respectively. According to an embodiment, the center pillar **210** can be affixed onto the base **210** first, follow by the corner pillars **220, 230, 240, 250**, follow by the middle pillars **260, 270, 280, 290**. Thus completing the construction of the bottom portion **200**. Because the middle pillars **260, 270, 280, 290** are installed last, they act to lock the remaining pillars in place. That is to say, once the middle pillars **260, 270, 280, 290** are successfully installed, all nine pillars are fixed in place and can no longer be removed.

In an embodiment, a first length of the center pillar **210** is different from a second length of the corner pillars **220, 230, 240, 250**, which is also different from a third length the middle pillars **260, 270, 280, 290**. Thus, when viewed from the side, as shown in FIG. 2, the pillars are staggered in different heights. The bundled tube configuration allows the resulting column **100** to have a higher structural integrity. In another embodiment, the lengths of all three types of pillars can be the same. In yet another embodiment, the length of two of the types of pillars can be the same, but different from the third type of pillar. For example, the center pillar **210** can be the same length as the corner pillars **220, 230, 240, 250**, but a different length as to the middle pillars **260, 270, 280, and 290**.

Referring to FIGS. 1 and 4, once the bottom portion **200** is constructed, addition pillars can be affixed on top of the pillars of the bottom portion **200**, forming the middle portion **300** of the column **100**.

The installation of the middle portion **300** is similar to the bottom portion **200**. That is, a center pillar of the middle portion **300** is affixed onto the center pillar **210** of the bottom portion **200** first. Follow by corner pillars of the middle portion **300** onto the corner pillars **220, 230, 240, 250** of the bottom portion **200** respectively. Lastly, middle pillars of the

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middle portion **300** are then dropped onto the middle pillars **260, 270, 280, 290** of the bottom portion **200** respectively.

Unlike the pillars of the bottom portion **200** however, in an embodiment, all the pillars of the middle portion **300** can have a same length. Thus, reducing manufacturing complexity and cost. However, the pillars of the middle portion **300** can have varying lengths when appropriate.

In an embodiment, the corner pillars of the middle portion **300** are twisted onto the corner pillars **220, 230, 240, 250** of the bottom portion **200** through the use of threaded stepped dowels that will be described in more detail later. The center pillar of the middle portion **300** can be twisted onto the center pillar of the bottom portion **200** through the use of a threaded stepped dowel, or it can be dropped onto a convention dowel or fastener protruding out of the center pillar **210** of the bottom portion **200**. Likewise, the middle pillars of the middle portion **300** are dropped onto dowels or fastener protruding out of the middle pillars **260, 270, 280, 290** of the bottom portion **200**.

To facilitate the embodiments where additional corner pillars are twisted onto corner pillars below, the corresponding center pillar can be octagonal in shape. Alternatively, the corresponding center pillar can be cylindrical. However, in the embodiments where additional corner pillars are dropped onto corner pillars below, the corresponding center pillar can be rectangular in shape, as no extra room is needed to allow for the twisting motion.

Because the pillars of the bottom portion **200** are staggered in heights, by placing addition pillars of the same length on top of the pillars of the bottom portion **200** would also result in staggered heights in the middle portion **300** as shown in FIG. 4. Again, the staggered configuration provides the resulting column **100** with better structural integrity, although non-staggered configuration can also be used to save cost of manufacturing.

Although FIGS. 1 and 4 show only one additional pillar on top of each of the pillars of the bottom portion **200**, additional pillars can be stacked on top to form an even longer middle portion **300**. That is to say, middle portion **300** can be as tall as necessary by affixing additional center pillars, corner pillars, and middle pillars on top of the pillars below. This is especially useful in a balloon construction where a column extends continuously from a sill plate to a top plate.

Once the middle portion **300** is at a desired height, the top portion **400** can then cap off the middle portion **300**, thus completing the construction of the column **100**. The top portion **400** can cap off the middle portion **300** in a variety of ways. For example, the top portion **400** can simply be a reverse of the bottom portion **200** comprising a center pillar, a plurality of corner pillars, and a plurality of middle pillars, where each pillar is affixed to the respective pillar of the middle portion **300** below.

Using FIG. 2 as an example, using corner pillars **220, 230, 240, and 250** of the bottom portion **200** as reference points, if the center pillar **210** of the bottom portion **200** is longer than the corner pillars of the bottom portion **200**, then the center pillar of the top portion **400** can be the shorter than corner pillars of the top portion **400** to compensate for the length of the center pillar of the bottom portion. Likewise, if the middle pillars of the bottom portion are shorter than the corner pillars of the bottom portion, then the middle pillars of the top portion **400** can be longer to compensate for the length difference, so that the constructed column has a flat top surface that can be affixed to other structural elements. However, it is to be understood that the top surface

of a column needs not be flat and column lengths of the top portion **400** can vary depending on the specific construction need.

Once each portion of the column **100** is construction, each individual pillar can further be reinforced from the sides through lateral fasteners such as additional dowels, nails, screws, or the like. Lateral reinforcement can also be metal rods or collars around the circumference of the column **100**.

In an embodiment, individual pillars can further be fastened to one another through a side fastener. For example, referring to FIG. 3, the corner pillar **220** can be fastened to the middle pillar **260** laterally through a dowel, a stepped dowel, a threaded stepped dowel, or other fasteners such as nails, toenails, screws, or the like. In this embodiment, the corner pillar **220** can be fastened to the middle pillar **260**, the corner pillar **230** can be fastened to the middle pillar **270**, the corner pillar **240** can be fastened to the middle pillar **280**, and the corner pillar **250** can be fastened to the middle pillar **290**. In addition, the middle pillars **260**, **270**, **280**, **290** can also be laterally fastened to the center pillar **210** through appropriate means such as dowels. When dowels are used to fasten adjoining pillars together laterally, each pillar can be predrilled for the dowels to permit for easy installation on-site.

Next, a threaded stepped dowel is described in more details herein. Referring to FIGS. 5A-6B, a threaded stepped dowel **500** according to an embodiment can be made of multiple sections, a base section **510** having the largest circumference or perimeter and each successive middle sections **520**, **530**, **540** having a smaller circumference or perimeter than the preceding section follow by a tip section **550**. When viewed in profile, such a dowel has a shape that is stepped or terraced, where the base section **510** is wider or larger than its opposed tip section **550**.

Although the figures illustrate three middle sections, the threaded stepped dowel **500** can comprise less or more middle sections. That is to say, the amount of middle section can range from one to as many as necessary. The threaded stepped dowel **500** can be made out of timber or engineered wood. Preferably, the threaded stepped dowel **500** is made out of the same material as the individual pillars of the bundled tube. The threaded stepped dowel **500** as described herein is suitable to be used to fasten components made of timber or engineered wood such as CLT or glulam. Certainly, the threaded stepped dowel **500** can also be used to fasten components not made of wood. In an embodiment, the threaded stepped dowel **500** can be one-way threaded. In yet another embodiment, the threaded stepped dowel **500** can be two-way threaded.

According to an embodiment, as shown in FIG. 6A, the sections **510**, **520**, **530**, **540**, **550** are generally cylindrical in shape with decreasing circumferences from one section to another. Each of the middle sections **520**, **530**, **540** further comprises grooves or threads **522**, **532**, **542** along the surface of these sections. The threads **522**, **532**, **542** spiral around each of their respective sections, forming a continuous thread in each section.

According to another embodiment, as shown in FIG. 6B, the middle sections **520**, **530**, **540** together with threads **522**, **532**, **542** form a generally sinusoidal profile. That is to say, when viewed from the side, each groove or thread curves inward toward a center axis of the threaded stepped dowels **500A**, whereas the sections between each thread curves outward away from the center axis of the threaded stepped dowels **500A**. The difference between the designs in FIG. 6A and FIG. 6B being that the additional curvatures as shown in the embodiment in FIG. 6B allows more exposures

of end-grain wood, thus allowing the threaded stepped dowel **500A** to absorb more moisture, whereas the flat profile of the threaded stepped dowel **500** as shown in FIG. 6A exposes more side-grain wood that absorbs less moisture. Nonetheless, both designs are suitable for use for the purpose of constructing the bundled tube as previously discussed.

In an embodiment, the threads **522**, **532**, **542** form a single-start threadform with respect to the corresponding middle sections **520**, **530**, **540**. Single-start refers to the configuration that each time the corresponding middle section is rotated by 360°, the middle section advanced axially by one ridge. However, when the threaded stepped dowel **500** comprises multiple middle sections, with each middle section having its own thread, the threaded stepped dowel **500** as a whole can be multi-start. For example, when the threaded stepped dowel **500** comprises three middle sections **520**, **530**, **540**, and each middle section having a corresponding thread **522**, **532**, **542**, although each middle section **520**, **530**, **540** is a single-start, the threaded stepped dowel **500** as a whole is a triple-start. That is to say, when the threaded stepped dowel **500** is rotated by one full rotation (360° degree), the threaded stepped dowel **500** advances by three ridges (one for each middle section). As it is to be appreciated, the stepped design of the threaded stepped dowel **500** increases the combined thread strength per rotation, i.e., being engaged to multiple additional ridges per rotation, instead of being engaged to one additional ridge per rotation.

Referring to FIG. 7, unlike a traditional dowel, the threaded stepped dowel **500** enables a first component **600** to be twisted onto a second component **700**, creating a more secured connection between the two components. Unlike a traditional dowel where the two components can be pulled apart through opposing forces, components fastened by the threaded stepped dowel **500** can only be disengaged by a reverse twisting motion, making the fastened components more difficult to disengage from one another.

In practice, the base section **510** of the threaded stepped dowel **500** is affixed onto the second component **700**. For example, the second component **700** can comprise a bore **710** on its top surface that corresponds to the size of the base section **510** so that the base section **510** can be inserted into or onto the second component **700**. Glue or other adhesives can also be applied to further secure the base section **510** within the bore **710** of the second component **700**.

On the flip side, the first component **600** can comprise a bore **610** at its bottom surface that mirrors the shape of the threaded stepped dowel **500**. That is to say, the bore **610** defines a cavity having generally the same shape as the profile of the threaded stepped dowel **500**, allowing the first component **600** to be able to twist onto the threaded stepped dowel **500**.

As illustrated in FIG. 7, the first component **600** can be one of the pillars of the column **100** as described before, and the second component **700** can be the base **202** as previously described. However, the same mechanism can also be used to connect one pillar with another. For example, the first component **600** can further comprise a bore **620** on its top surface that can be used to engage with another threaded stepped dowel, which then engages with another pillar on top.

Referring to FIG. 8A, an alternative design of the threaded stepped dowel **500** as discussed prior can be a double-sided threaded stepped dowel **800**. Mechanically, this alternative design is largely similar to the threaded stepped dowel **500**.

In this embodiment, a base section **810** is located toward the center of the double-sided threaded stepped dowel **800** with middle sections extending outward therefrom. Similar to the threaded stepped dowel **500**, each successive section of the double-sided threaded stepped dowel **800** decreases in circumference with the tip sections **880**, **890** having the smallest circumferences of all the sections. Likewise, each middle sections **820**, **830**, **840**, **850**, **860**, **870** is provided with thread or groove that spirals around the outer surface of each section. Similar to a threaded stepped dowel **500**, the double-sided threaded stepped dowel **800** can have any number of middle sections. Moreover, the number of the middle sections on one side of the double-sided threaded stepped dowel **800** need not be the same as the number of the middle sections on the other side. In yet another embodiment, the base section **810** can be omitted altogether, as shown in FIG. **8B**.

FIG. **9** shows another embodiment of one of the pillars to be used for the bundled tube in forming column **100**. In this embodiment, a pillar **900** can contain bores **910**, **920** on the top and bottom surfaces of the pillar **900**. The bores **910**, **920** each defining a cavity to each accept a threaded stepped dowel. The pillar **900** can be used as one of the corner pillars as discussed previously especially in the middle portion **300** of the column **100**. Alternatively, the pillar **900** can be octagonal or cylindrical in shape, which then can be used as one of the center pillars.

FIG. **10** illustrates an example of how a threaded stepped dowel can be used when platform construction is desired. A base section of a threaded stepped dowel **1010** can be installed onto a first component **1020**. The first component **1020** can be various panels typically found in construction of a building, such as a floor panel, a ceiling panel, or a wall panel. In an embodiment, only the base section of the threaded stepped dowel **1010** is contained within the first component **1020**. In another embodiment, the base section, together with one or more middle sections of the threaded stepped dowel **1010** can be contained within the first component **1020**. For example, in FIG. **10**, the base section, together with one middle section is contained within the first component **1020**. Thus, it is understood that so long as enough middle section(s) of the threaded step dowel **1010** remain available to mate with a second component, the exact configuration can be adjusted as necessary.

As before, a second component **1030** can be affixed onto the first component **1020** by twisting the second component **1030** onto the threaded stepped dowel **1010**. Specifically, the second component **1030** can comprise a corresponding bore **1040** that mates with the threaded stepped dowel **1010**. The second component **1030** can be a column, a beam, a pillar, or a post. It is to be appreciated that the threaded stepped dowel **1010** can also be installed on the second component **1030** instead of or in addition to the first component **1020**. By way of example, a threaded stepped dowel can be installed on a beam or column, and a panel can be affixed onto the beam therefrom.

Certainly, in a platform construction, more than one pillars can be affixed onto a panel. In a typical construction, about 25 pillars can be used per floor. These 25 pillars can all be affixed to the floor panel using threaded stepped dowels. Illustratively, assuming a threaded stepped dowel is 4.5 inches in diameter. Said threaded stepped dowel would be able to withstand a crushing weight of about 80,000 pounds ($2.25^2 \cdot \pi \cdot 5,000 \approx 80,000$). Thus, when 25 of these threaded stepped dowels are used, the crushing weight that these threaded stepped dowels can withstand is about 2 million pounds, well enough for a typical building.

Specific embodiments of a column formed by bundled tube and a threaded stepped dowel according to the present invention have been described for the purpose of illustrating the manner in which the invention can be made and used. It should be understood that the implementation of other variations and modifications of this invention and its different aspects will be apparent to one skilled in the art, and that this invention is not limited by the specific embodiments described. Features described in one embodiment can be implemented in other embodiments. The subject disclosure is understood to encompass the present invention and any and all modifications, variations, or equivalents that fall within the spirit and scope of the basic underlying principles disclosed and claimed herein.

What is claimed is:

1. A system for construction comprising:

a threaded stepped dowel comprising:

- a base section having a first circumference;
- a first middle section having a first middle circumference smaller than the first circumference;
- a second middle section having a second middle circumference smaller than the first middle circumference; and
- a tip section having a third circumference smaller than the second middle circumference,

wherein each of the first and the second middle sections are generally cylindrical in shape and include a threaded outer surface,

wherein a continuous thread spirals around each of the first and the second middle sections, and

wherein the tip section is a cylinder that includes a smooth outer surface;

a first component having a first bore for receiving the first and the second middle sections and the tip section of the threaded stepped dowel; and

a second component having a second bore for receiving the base section of the threaded stepped dowel;

wherein the base section of the threaded stepped dowel is affixed into the second bore, and

wherein the first component is attached to the second component through twisting the first component onto the threaded stepped dowel.

2. The system for construction of claim 1, wherein the first component and the second component are each a panel or a pillar.

3. The system for construction of claim 1, wherein the first bore is a cavity having a shape that corresponds to a profile of the threaded stepped dowel.

4. The system for construction of claim 1, wherein the first component can only be disengaged from the second component through a twisting motion.

5. The system for construction of claim 1, wherein the first bore is provided on a bottom surface of the first component, and

wherein the first component further includes a third bore on a top surface for receiving another threaded stepped dowel.

6. The system for construction of claim 1, wherein the first component, the second component, and the threaded stepped dowel are made of a same material.

7. A system for construction comprising:

a threaded stepped dowel comprising:

- a first middle section having a first circumference;
- a second middle section having a second circumference;

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a first tip section adjoining the first middle section, the first tip section having a circumference smaller than the first circumference; and

a second tip section adjoining the second middle section, the second tip section having a circumference smaller than the second circumference,

wherein the first middle section and the second middle section are each generally cylindrical in shape with a threaded outer surface that includes a thread, the thread comprising alternating outer portions and round recess portions immediately following one another,

wherein the first tip section and the second tip section are each a cylinder that includes a smooth outer surface,

wherein the first middle section further comprises a first plurality of sections, with each successive section having a smaller circumference than the last, and each of the first plurality of sections is generally cylindrical in shape with a threaded outer surface that includes a thread, and

wherein the second middle section further comprises a second plurality of sections, with each successive section having a smaller circumference than the last, and each of the second plurality of sections is generally cylindrical in shape with a threaded outer surface that includes a thread;

a first component having a first bore for receiving the first middle section and the first tip section of the threaded stepped dowel; and

a second component having a second bore for receiving the second middle section and the second tip section of the threaded stepped dowel;

wherein the threaded stepped dowel is attached to the second component through twisting the threaded stepped dowel onto the second component, and

wherein the first component is attached to the second component through twisting the first component onto the threaded stepped dowel.

8. The system for construction of claim 7, wherein the first component and the second component are each a panel or a pillar.

9. The system for construction of claim 7, wherein the first bore is a cavity having a shape that corresponds to a profile of the threaded stepped dowel.

10. The system for construction of claim 7, wherein the first component can only be disengaged from the second component through a twisting motion.

11. The system for construction of claim 7, wherein the first bore is provided on a bottom surface of the first component, and

wherein the first component further includes a third bore on a top surface for receiving another threaded stepped dowel.

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12. The system for construction of claim 7, wherein the first component, the second component, and the threaded stepped dowel are made of a same material.

13. A method for construction comprising:

providing a first component having a first bore for receiving a first portion of a threaded stepped dowel;

providing a second component having a second bore for receiving a second portion of the threaded stepped dowel;

attaching the threaded stepped dowel to the second component; and

twisting the first component onto the threaded stepped dowel, thereby attaching the first component to the second component,

wherein the first bore defines a cavity having a threaded stepped shape corresponding to a threaded stepped profile of the first portion, and

wherein the threaded stepped dowel is configured to withstand a crushing weight of about 80,000 pounds.

14. The method for construction of claim 13 further comprising:

detaching the first component from the second component through a twisting motion.

15. The method for construction of claim 13, wherein the attaching the threaded stepped dowel to the second component comprises twisting the threaded stepped dowel into the second bore of the second component.

16. The method for construction of claim 13, wherein the threaded stepped dowel comprises:

a base section having a first circumference;

a middle section having a second circumference smaller than the first circumference; and

a tip section having a third circumference smaller than the second circumference,

wherein the middle section is generally cylindrical in shape with a threaded outer surface that includes a thread.

17. The method for construction of claim 13, wherein the threaded stepped dowel comprises:

a first middle section having a first circumference;

a second middle section having a second circumference;

a first tip section adjoining the first middle section, the first tip section having a circumference smaller than the first circumference; and

a second tip section adjoining the second middle section, the second tip section having a circumference smaller than the second circumference,

wherein the first middle section and the second middle section are each generally cylindrical in shape with a threaded outer surface that includes a thread, and

wherein the first tip section and the second tip section are each a cylinder that includes a smooth outer surface.

18. The method for construction of claim 13, wherein the first component, the second component, and the threaded stepped dowel are made of a same material.

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